

Food Waste Reduction with Agile Circular Supply Chains

Abstract

Food waste in retail supply chains causes economic and environmental losses. This paper studies how agile project management and circular economy principles can be combined to reduce food waste. We run a twelve-week pilot with three regional retailers using two-week sprints, demand forecasting and redistribution of surplus food to food banks. Waste fell by 23 percent compared to a control group.

1 Introduction

Roughly one third of food produced is lost or wasted. Traditional waterfall planning in supply chains reacts slowly to demand changes. Agile methods such as Scrum emphasise short feedback loops, which may help perishable inventory decisions. Circular economy thinking treats surplus as a resource rather than waste.

2 Methodology

We used a quasi-experimental design. Three stores adopted sprint-based replenishment planning with daily stand-ups between store managers and logistics planners. Demand forecasting used gradient boosted trees trained on two years of point-of-sale data. Surplus items were routed to partner food banks and composting facilities.

3 Results

Pilot stores reduced food waste by 23 percent and stock-outs by 6 percent. Forecast error decreased from 18 to 12 percent MAPE. Staff reported that sprint reviews improved coordination between purchasing and store operations.

4 Conclusion

Combining agile project management with circular economy practices measurably reduces retail food waste. Limitations include the small sample of three stores and a short pilot. Future work should test the approach across seasons and in other countries.

References

[1] Schwaber, K. Agile Project Management with Scrum. 2004.